

Integration by Parts: Choosing the Pieces

Answer Key

AP Calculus AB/BC

Quick Answers

1. $xe^x - e^x$

2. $x \ln(x) - x$

3. $\sin(x) - x \cos(x)$

4. $\frac{x^2 \ln(x)}{2} - \frac{x^2}{4}$

5. $\frac{e^2+1}{4}$

6. (a) antiderivative = $x^2e^x - 2xe^x + 2e^x$, —

7. $x \operatorname{atan}(x) - \frac{\ln(1+x^2)}{2}$

8. $\frac{e^x(\sin(x) - \cos(x))}{2}$

9. -2

10. (a) value on $[0, 3] = 1 - \frac{4}{e^3}$, (b) limit = 1, —

What is verified

11 of 13 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 6, 10. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

FUN-6.E – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 13 tagged checks.

1. $\int xe^x dx$.

Solution. $u = x$, $dv = e^x dx$, so $du = dx$ and $v = e^x$:

$$\int xe^x dx = xe^x - \int e^x dx.$$

$$\boxed{xe^x - e^x + C}$$

Differentiating $xe^x - e^x$ gives $e^x + xe^x - e^x = xe^x$, as it must.

2. $\int \ln x dx$.

Solution. There is only one factor, so take $u = \ln x$ and $dv = dx$; then $du = \frac{dx}{x}$ and $v = x$:

$$\int \ln x dx = x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - \int dx.$$

$$\boxed{x \ln x - x + C}$$

3. $\int x \sin x \, dx$.

Solution. $u = x$, $dv = \sin x \, dx$, so $v = -\cos x$:

$$\int x \sin x \, dx = -x \cos x + \int \cos x \, dx.$$

$$\boxed{\sin x - x \cos x + C}$$

4. $\int x \ln x \, dx$.

Solution. The logarithm is the factor that improves when differentiated, so $u = \ln x$ and $dv = x \, dx$, giving $v = \frac{x^2}{2}$:

$$\int x \ln x \, dx = \frac{x^2 \ln x}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} \, dx = \frac{x^2 \ln x}{2} - \frac{1}{2} \int x \, dx.$$

$$\boxed{\frac{x^2 \ln x}{2} - \frac{x^2}{4} + C}$$

5. $\int_0^1 x e^{2x} \, dx$.

Solution. $u = x$, $dv = e^{2x} \, dx$, so $v = \frac{e^{2x}}{2}$:

$$\left[\frac{x e^{2x}}{2} \right]_0^1 - \frac{1}{2} \int_0^1 e^{2x} \, dx = \frac{e^2}{2} - \left[\frac{e^{2x}}{4} \right]_0^1 = \frac{e^2}{2} - \frac{e^2}{4} + \frac{1}{4}.$$

$$\boxed{\frac{e^2 + 1}{4}}$$

6. $\int x^2 e^x \, dx$. (a) Evaluate. (b) Why twice?

Solution (a). First pass: $u = x^2$, $dv = e^x \, dx$, giving $x^2 e^x - 2 \int x e^x \, dx$. The remaining integral is problem 1, so

$$\int x^2 e^x \, dx = x^2 e^x - 2(x e^x - e^x).$$

$$\boxed{x^2 e^x - 2x e^x + 2e^x + C}$$

Solution (b) — instructor-judged. Each application of parts differentiates the polynomial factor once, lowering its degree by one: $x^2 \rightarrow 2x \rightarrow 2$. After the first pass the leftover integral still contains an x , so it is the same shape one power down; after the second pass the polynomial is a constant and the integral is elementary. Full credit needs that degree-lowering idea, not just “the first pass didn’t finish”.

7. $\int \arctan x \, dx$.

Solution. $u = \arctan x$, $dv = dx$, so $du = \frac{dx}{1+x^2}$ and $v = x$:

$$\int \arctan x \, dx = x \arctan x - \int \frac{x}{1+x^2} \, dx,$$

and the leftover integral is a substitution, $w = 1 + x^2$, giving $\frac{1}{2} \ln(1 + x^2)$.

$$\boxed{x \arctan x - \frac{\ln(1+x^2)}{2} + C}$$

8. $\int e^x \sin x \, dx$.

Solution. Call the integral I . With $u = \sin x$, $dv = e^x dx$: $I = e^x \sin x - \int e^x \cos x \, dx$. Applying parts again to the new integral (same choice of dv) gives $\int e^x \cos x \, dx = e^x \cos x + I$, so

$$I = e^x \sin x - e^x \cos x - I \implies 2I = e^x(\sin x - \cos x).$$

$$\boxed{\frac{e^x(\sin x - \cos x)}{2} + C}$$

The integral came back unchanged, so it could be solved for algebraically — that is the whole trick here.

9. $\int_0^\pi x \cos x \, dx$.

Solution. $u = x$, $dv = \cos x \, dx$, so $v = \sin x$:

$$[x \sin x]_0^\pi - \int_0^\pi \sin x \, dx = 0 + [\cos x]_0^\pi = -1 - 1.$$

$$\boxed{-2}$$

10. $\int_0^b x e^{-x} \, dx$. (a) $b = 3$. (b) The limit. (c) The strategic choice.

Solution (a). $u = x$, $dv = e^{-x} dx$, so $v = -e^{-x}$:

$$[-x e^{-x}]_0^b + \int_0^b e^{-x} \, dx = -b e^{-b} + [-e^{-x}]_0^b = 1 - (b+1)e^{-b}.$$

At $b = 3$ this is

$$\boxed{1 - 4e^{-3}}$$

Solution (b). As $b \rightarrow \infty$ the exponential beats the linear factor, so $(b+1)e^{-b} \rightarrow 0$:

$$\boxed{1}$$

Solution (c) — instructor-judged. Differentiating x removes it completely after one pass, while e^{-x} integrates back to itself (up to a sign) and never becomes harder. Swapping the roles would differentiate the exponential — which stays exponential — and integrate x into $\frac{x^2}{2}$, leaving an integral worse than the original. Full credit needs the reason, not just the correct assignment.